Modernizing ops_pipeline3

From a Data Factory pipeline to a Spark notebook in Microsoft Fabric

Prepared for the customer · 2026-09-03

Chapter 1 – The `ops_pipeline3` pipeline

Purpose. `ops_pipeline3` is a **metadata-driven incremental ingestion** pipeline. It copies data from an **Eventhouse (KQL database)** into **Lakehouse Delta tables**, driven by a control table so that new source tables can be onboarded by simply adding a row – no change to the pipeline itself.

Parameters & variables

- `ops_lakehouse` – the target Lakehouse (holds the control table and the destination tables).
- `ops_kql_db` – the source KQL / Eventhouse database.
- `vRunId` – a variable capturing the pipeline Run Id.

Step-by-step

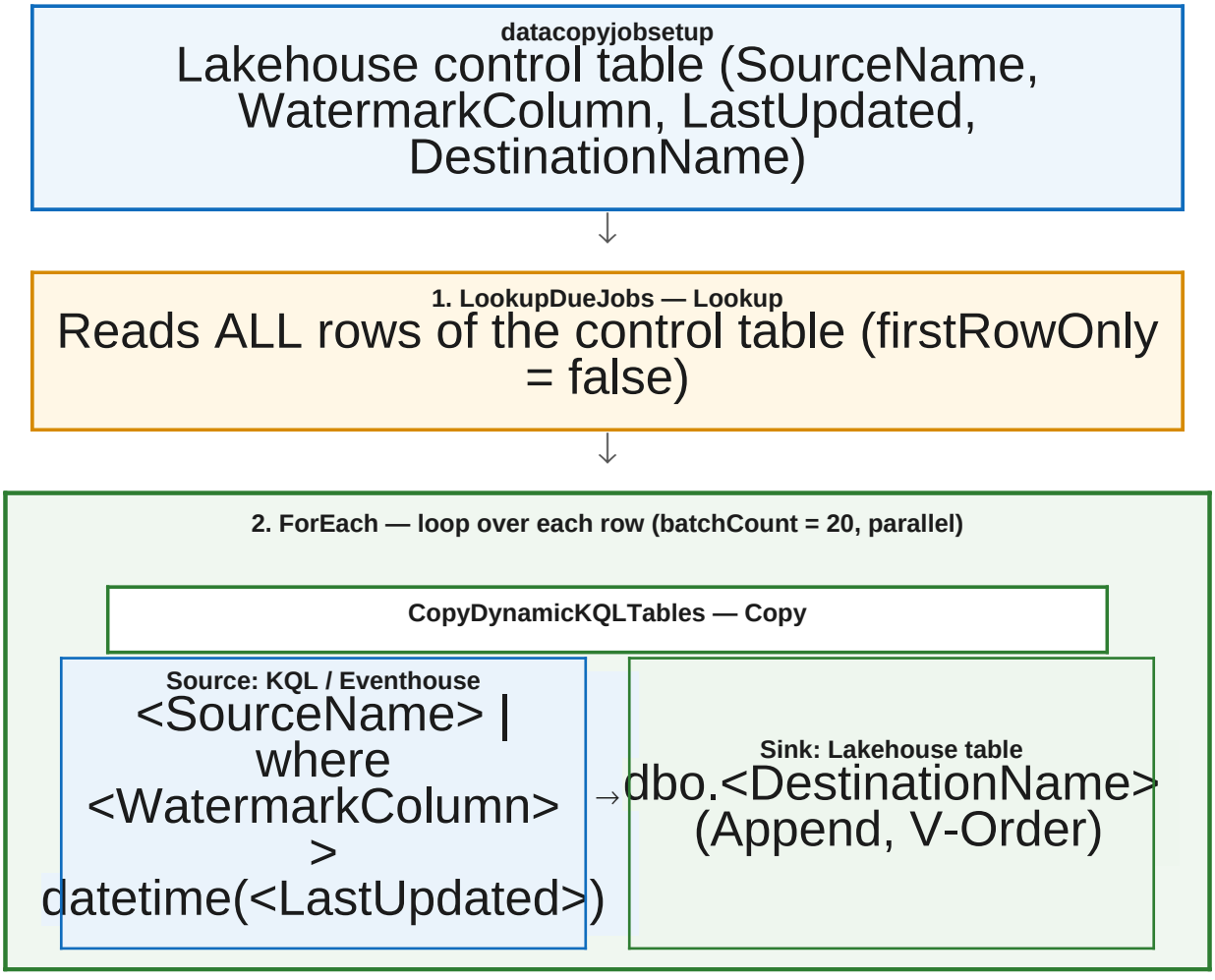
1. LookupDueJobs (Lookup). Reads **every row** of the control table `dbo.datacopyjobsetup` in the Lakehouse (`firstRowOnly = false`). Each row describes one copy job with the columns `SourceName`, `WatermarkColumn`, `LastUpdated` and `DestinationName`.

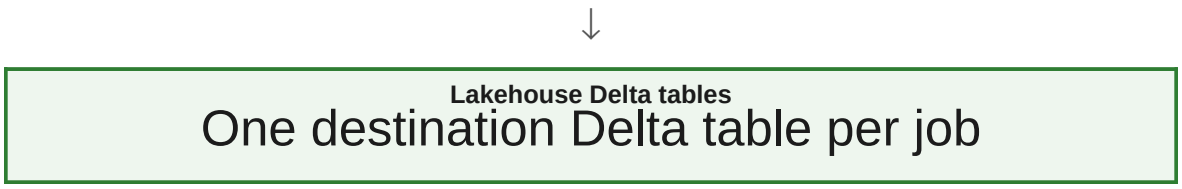
2. ForEach (loop, batchCount = 20). Iterates over the rows returned by the lookup, processing up to 20 in parallel. For each row it runs one Copy activity:

CopyDynamicKQLTables (Copy). Source = a **dynamic KQL query** against the Eventhouse: `<SourceName> | where <WatermarkColumn> > datetime(<LastUpdated>)`, i.e. only rows newer than the stored watermark (an incremental pull). **Sink** = the Lakehouse Delta table `dbo.<DestinationName>` in **Append** mode (with V-Order).

> Note: the pipeline itself never updates `LastUpdated`. The watermark must be maintained elsewhere – the notebook in Chapter 2 offers an optional step to advance it.

Graphical representation




Lakehouse Delta tables
One destination Delta table per job

Chapter 2 – The replacement notebook

The following notebook reproduces the pipeline in PySpark. Markdown cells are shown as text; code cells are rendered with the same syntax highlighting you see in the Fabric notebook editor.

Replace `ops_pipeline3` with Spark

Metadata-driven incremental ingestion from an Eventhouse (KQL database) into Lakehouse Delta tables.

This notebook replaces the pipeline's two steps:

1. **LookupDueJobs** → read the control table `datacopyjobsetup`.
2. **ForEach + CopyDynamicKQLTables** → for each job, run an incremental KQL query and append the result to `<DestinationName>`.

Each control row provides: `SourceName`, `WatermarkColumn`, `LastUpdated`, `DestinationName`.

The KQL query built per job is: `<SourceName> | where <WatermarkColumn> > datetime(<LastUpdated>)`.

> Tables are accessed by their **absolute OneLake path**, so no lakehouse needs to be attached to the notebook.

Parameters

```
# Eventhouse / KQL source (pipeline parameter ops_kql_db). kql_cluster =
"https://trd-7qm6ccfqm2rr8uzwff.z0.kusto.fabric.microsoft.com" kql_database =
"98405e07-5da7-48bd-a7c4-4fbfd276d4d1" # KQL database UID (GUID), not the display name # Lakehouse holding
the control table and destination tables (pipeline parameter ops_lakehouse). workspace_id =
"914d92b7-d202-4390-a2e2-0d7c3fcb3483" lakehouse_id = "33e1f298-b3f3-4a67-ad30-732cb798d525" #
ops_lakehouse artifact (GUID) config_table = "datacopyjobsetup" # control table read by LookupDueJobs
dest_schema = "dbo" # schema in a schema-enabled lakehouse; set "" if not schema-enabled max_parallel = 1
# sequential: Eventhouse export capacity is 1 (one .export at a time) advance_watermark = False # pipeline
does NOT advance the watermark; keep off to match it
```

Setup: OneLake paths, Kusto token, helpers

```
from concurrent.futures import ThreadPoolExecutor, as_completed import time import notebookutils
_TABLES_ROOT = f"abfss://{workspace_id}@onelake.dfs.fabric.microsoft.com/{lakehouse_id}/Tables" def
table_path(table_name: str, schema: str = None) -> str: """Absolute OneLake path to a Delta table (works
without an attached lakehouse)."""  schema = dest_schema if schema is None else schema return
f"{_TABLES_ROOT}/{schema}/{table_name}" if schema else f"{_TABLES_ROOT}/{table_name}" # AAD token for the
Eventhouse/Kusto cluster, using the notebook's identity. kusto_token =
notebookutils.credentials.getToken(kql_cluster) def read_kql(query: str): """Run a KQL query against the
Eventhouse and return a Spark DataFrame."""  return (
spark.read.format("com.microsoft.kusto.spark.datasource") .option("kustoCluster", kql_cluster)
.option("kustoDatabase", kql_database) .option("kustoQuery", query) .option("accessToken", kusto_token)
.option("readMode", "ForceDistributedMode") # export-based read, no 500K-row query cap .load() ) def
build_query(source_name: str, watermark_column: str, last_updated) -> str: """Reproduce the pipeline's
dynamic KQL: <source> | where <col> > datetime(<iso>)."""  iso =
last_updated.strftime("%Y-%m-%dT%H:%M:%SZ") if last_updated is not None else "1900-01-01T00:00:00Z" return
f"{source_name} | where {watermark_column} > datetime({iso})"
```

Step 1 — LookupDueJobs (read the control table)

```
jobs = spark.read.format("delta").load(table_path(config_table)).collect() print(f"Loaded {len(jobs)} copy
job(s) from {table_path(config_table)}") for j in jobs: print(f" - {j['SourceName']} ->
{j['DestinationName']} (watermark {j['WatermarkColumn']} > {j['LastUpdated']})")
```

Step 2 — ForEach + Copy (incremental KQL → Lakehouse append)

```
def run_job(job) -> dict: source_name = job["SourceName"] watermark_column = job["WatermarkColumn"]
last_updated = job["LastUpdated"] destination = job["DestinationName"] query = build_query(source_name,
watermark_column, last_updated) last_err = None for attempt in range(5): try: df = read_kql(query)
row_count = df.count() if row_count > 0: ( df.write.mode("append") .format("delta") .option("mergeSchema",
"true") .save(table_path(destination)) ) return {"source": source_name, "destination": destination,
"rows": row_count, "query": query} except Exception as exc: # noqa: BLE001 - retry transient Kusto
throttling last_err = exc if "Throttled" in str(exc) or "throttling" in str(exc): time.sleep(min(60, 5 * 2
** attempt)) # exponential backoff continue raise raise last_err results = [] with
ThreadPoolExecutor(max_workers=max_parallel) as pool: futures = {pool.submit(run_job, j): j for j in jobs}
for fut in as_completed(futures): job = futures[fut] try: res = fut.result() print(f"OK {res['source']} ->
{res['destination']}: {res['rows']} row(s) appended") results.append(res) except Exception as exc: # noqa:
BLE001 - report per-job failure, continue others print(f"FAIL {job['SourceName']} ->
{job['DestinationName']}: {exc}") results.append({"source": job["SourceName"], "destination":
job["DestinationName"], "error": str(exc)}) print(f"\nCompleted {len(results)} job(s).")
```

Optional — advance the watermark

The original pipeline never updated `LastUpdated`, so re-runs re-copy the same rows. Enable this to make the run truly incremental by advancing each job's watermark to the max value just ingested.

```
from pyspark.sql import functions as F if advance_watermark: from delta.tables import DeltaTable control =
DeltaTable.forPath(spark, table_path(config_table)) for res in results: if res.get("rows", 0) and "error"
not in res: job = next(j for j in jobs if j["DestinationName"] == res["destination"]) new_max = (
spark.read.format("delta").load(table_path(res["destination"]))
.agg(F.max(job["WatermarkColumn"]).alias("m")) .collect()[0]["m"] ) if new_max is not None:
control.update( condition=F.col("DestinationName") == res["destination"], set={"LastUpdated":
F.lit(new_max)}, ) print(f"Watermark for {res['destination']} advanced to {new_max}") else:
print("advance_watermark is False — matching original pipeline behavior (watermark unchanged).")
```

Chapter 3 – Parallelism & scaling

The pipeline's ForEach used **batchCount = 20**. When translated to Spark, the real limit is not the notebook – it is how much the **Eventhouse** can serve concurrently. Two reader modes exist, each with a different constraint:

- **Single (query) mode** – reads through the query API. Simple, but capped at **500,000 rows** per query. Good only for small tables.
- **Distributed mode** – reads via `.export` to storage. **No row cap** (used for the large `Measurements` tables), but each export consumes an **export slot**.

Why concurrency was limited to 1

The Eventhouse reported `Export Capacity: 1` – only one `.export` may run at a time. Running 8 jobs in parallel therefore got **throttled**. The number of concurrent exports scales with the compute:

concurrent exports ≈ total cores × 0.75

So to allow `max_parallel = 8` you need roughly **8 ÷ 0.75 ≈ 11 warm cores**.

Scaling on your F64 capacity

F64 has ample headroom; the lever is **how many Eventhouse cores are kept warm**. In the Fabric portal: open the **Eventhouse** → top toolbar **Capacity Planner** (currently Off) and raise the **Minimum consumption** level. Verify with the KQL command `.show capacity` (look at the **Export** row) before increasing `max_parallel`.

Desired max_parallel	Warm cores (~)	Action
1 (current)	any	nothing – sequential, safe on Capacity 1
4	~6	raise Minimum consumption one–two levels
8	~11	Minimum consumption to a Medium/Large level

> Trade-off: keeping more cores warm increases cost. Raise it for the batch window, then lower it again. The notebook keeps a retry-with-backoff as a safety net for transient throttling.